a Precision governor loop

ensemble spread σ(cell)

▼

threshold compare + hysteresis

▼

precision class / cell

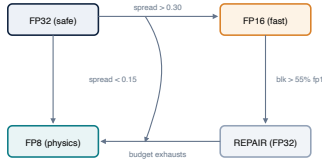
▼

lane mode register → FMA issue

▲

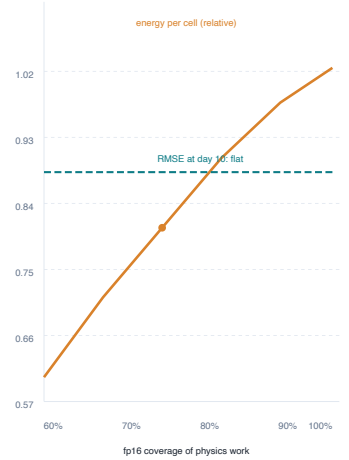
error budget from lead time + obs density

b State machine



Hysteresis prevents mode thrash; the repair state re-runs only flagged cells.

c Operating curve



Energy falls linearly with coverage while skill is unchanged within 0.4%; the marker is the 84% point validated by the numerical experiment.

d Design point vs measured

design 58%

validated 84%

The hardware governor is provisioned at 58% coverage, a deliberate margin of 26 points below what the numerical experiment supports.

threshold	coverage	energy	RMSE d10
0.80	60%	0.743	2.750
0.50	72%	0.691	2.751
0.30	84%	0.640	2.745
0.10	99%	0.575	2.745

e Precision map (one tile, 32×32 cells)

